

Part 1: Architectural Design Specification

1.1 Introduction

Broken Windows Theory was first proposed by Wilson and Kelling [1], who suggest that neglect of the built environment, such as graffiti, litter and derelict buildings creates an impression of urban decay that leads to more serious crime. This research will test the theory empirically to determine if there is a correlation between environmental neglect and crime in London by using fly-tipping to measure neglect and recycling rates to measure community engagement. A complete cloud-based data analytics pipeline is implemented in Microsoft Azure, using Databricks and Tableau.

The empirical test of this theory will use the amount of recycling and fly-tipping to measure community engagement and environmental neglect, respectively, to see if there is a correlation between environmental neglect and crime in London. Fly-tipping is a direct measure of visible physical disorder and household recycling rates act as an indicator of active citizen participation and collective responsibility. The intention of this study is to provide a full assessment of the Broken Windows Theory in relation to the London boroughs and the overall borough-based crime rates.

The aim is to measure the robustness of association between environmental deterioration and crime rates by London boroughs and to see if this is reduced by the involvement of the community. The results will be fed into the current academic discussion and could be used to support evidence-based policies for urban management and crime prevention.

Borough was selected as the unit of analysis because it is the finest spatial grain at which all three datasets are simultaneously and consistently available across full study window. Lower Super Output Area (LSOA) resolution would have provided greater spatial sensitivity and reduced ecological fallacy risk; however, neither the fly-tipping nor the recycling series is published at LSOA granularity, and disaggregating via population weighting would have introduced synthetic variance. Borough therefore represents the highest-resolution unit defensible from the data and aligns with the administrative level at which London crime-prevention policy is commissioned.

The entire data analytics pipeline is realized in the cloud, with a combination of Databricks and Tableau. The architecture is based on a medallion approach and is layered to ensure data quality and auditability in bronze, silver and gold layers. Three datasets from the London Datastore are analyzed: borough-level data on crime committed by the Metropolitan Police Service; data on fly tipping; and data on household waste recycling rates. These relationships can be quantified through correlation analysis and linear modelling and are used to provide empirical evidence to assess Broken Windows Theory in London.

1.2 Overall Architecture

The architecture pipeline is a cloud-native data Lakehouse architecture that utilizes Azure Data Lake Storage Gen2 [7], Databricks, Delta Lake and Tableau.

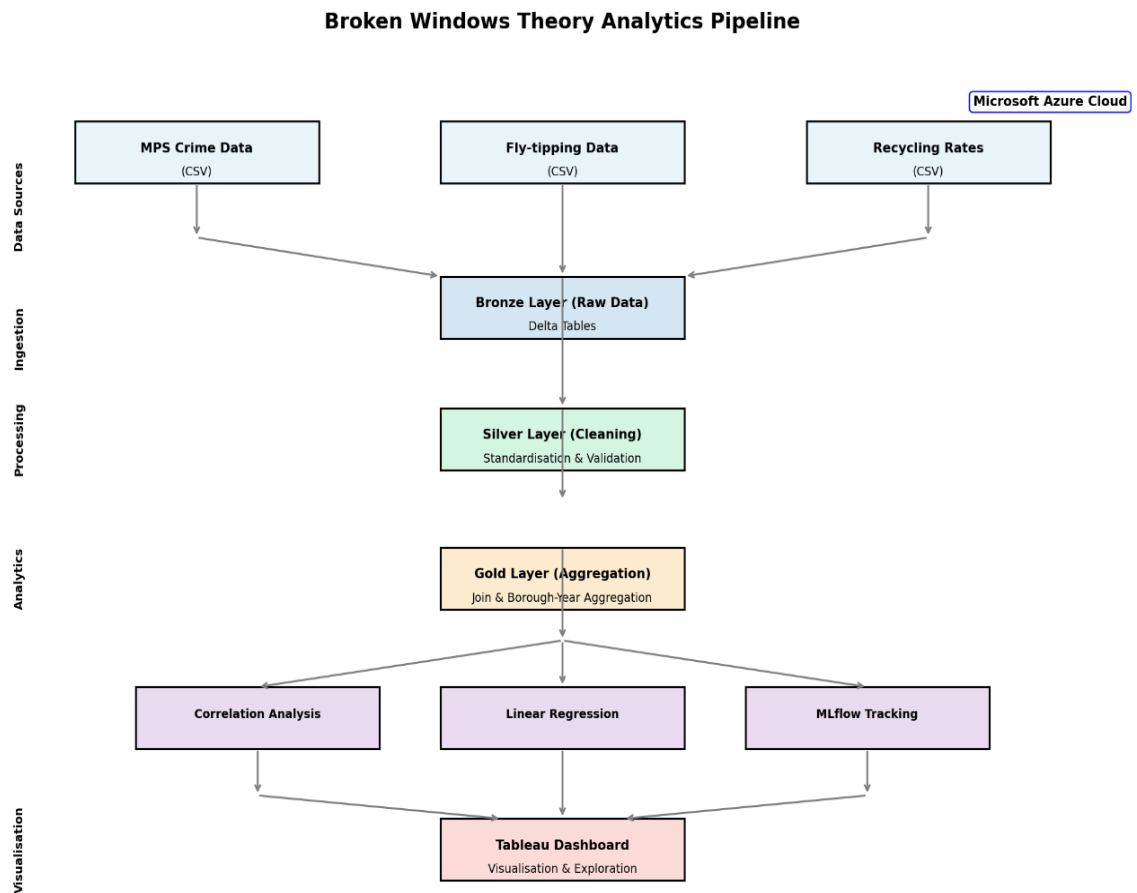


Figure 1: Broken Windows Theory Analytics Pipeline Architecture

Data Flow: CSV → Azure Data Lake - Bronze → Databricks - Silver/Gold → Delta Lake → Tableau

This architecture was selected for three principal reasons. Azure cloud services enables the crime data set, with more than one million records, to be processed without loss of performance, thereby achieving scalability. Data integration is achieved by joining and combining heterogeneous crime data, fly-tipping data and recycling data in a seamless way, despite their varying formats and structures. Support for batch and incremental processing ensures the suitability of urban analytics in an urban context, and the continuity of historical information for longitudinal analysis for over 10 years.

A cloud-native architecture was preferred to local processing because the pipeline must remain available for re-execution and audit beyond the lifespan of any single workstation, and the ACID transactional storage guarantees underpinning the medallion design are impractical to replicate in a local file system. Local processing with Python and CSV files would have been sufficient for a one-off correlation calculation but could not have supported the reproducibility, auditability and scalability requirements that distinguish an analytics-engineering deliverable from an ad hoc analysis script.

This flow is supported by the medallion pattern. Raw CSV files are stored in a 'Bronze layer' that will not change, ensuring they can be reproduced and audited. Silver layer cleans data such as date parsing, removing commas from numeric strings and borough name standardization. Gold layer performs aggregation at the borough-year level and performs a join on all 3 datasets. ACID transaction support, schema enforcement and time travel are all capabilities that Delta Lake supports and is necessary in production data pipelines. The last layer of gold is used by Tableau for interactive dashboard

visualization, allowing stakeholders to examine the relationships between crime, fly-tipping and recycling rates by borough and time, across London.

The medallion architecture was preferred over a flat lake or a traditional warehouse schema because it imposes progressive, auditable quality gates between raw and analytical layers while preserving source-data immutability. A flat-lake design would have collapsed all transformations into a single step, making the provenance of any reported figure impossible to reconstruct independently. A traditional warehouse schema would have required up-front dimensional modelling decisions that the exploratory nature of this study did not warrant and would have lacked the schema-flexibility needed when integrating three heterogeneous, community-published datasets whose column structures differ materially.

1.3 Data Sources

Crime Dataset MPS Borough-Level Crime: Total crime by borough by month, operationalising the crime component of Broken Windows Theory. This includes coverage of the 32 London boroughs and the City of London, and data from 2008-2023. The data is susceptible to masking within-borough differences and only reporting incidents [2].

Total crime was used as the dependent variable because Broken Windows Theory advances a claim about the overall criminal character of a place, not about a specific offence type. Restricting the analysis to a subcategory such as anti-social behavior would have introduced circularity, since such offences are conceptually proximate to the visible-disorder construct. Total crime provides a conservative test: if the predicted association is genuine, it should be detectable in the aggregate rather than only in a selectively chosen subcategory.

Fly-tipping Dataset: This dataset operationalises the concept of visible environmental neglect by measuring fly-tipping. Advantages include aggregation at the same borough level as crime, and annual data. Weaknesses include only capturing reported fly-tipping and varying enforcement efforts across boroughs [2].

Household Recycling Rates Dataset: This data set provides a measure of community participation and solidarity, with increasing recycling rates as a measure of citizen engagement. This dataset offers consistent measurement across all London boroughs and historical data from 1998 to 2024. Weaknesses include the potential shaping of the data by policies of the local council [2].

Household recycling rate was selected as the third dataset and community-engagement proxy because it captures repeated, observable civic behavior requiring sustained co-operation between residents and local services, consistent with the collective-efficacy construct invoked by the community arm of Broken Windows Theory. Alternative proxies such as voter turnout were rejected because they are dominated by national political determinants; third-sector membership density was unavailable at borough granularity for the full study window. Recycling participants is acknowledged to be partly shaped by local-authority collection policy, which is documented as a limitation in 3.5.

1.4 CSV Ingestion - Bronze Layer

Data ingestion was performed into Azure Data Lake Storage Gen2 using Databricks' built-in CSV reader with auto-detect schema. The bronze layer represents a landing zone for raw data, which is stored in its original format. Preserving raw data provides reproducibility and auditability [3]. Retaining the original CSV files allows any issue with the processing to be traced back to source without re-ingestion and full auditability.

1.5 Storage Design

Azure Data Lake Storage Gen2 offers a hierarchical namespace with three zones, in a medallion architecture pattern [4]. The bronze zone contains raw CSV files. The silver layer stores transformed data in Delta format. The gold layer stores analytical data in Delta format.

We chose Delta for three reasons. Schema enforcement guarantees that data is correct by rejecting writes that are incompatible with the expected schema. ACID transactions allow for concurrent reads and write. Time travel enables querying past versions of the data, aiding with debugging and reproducibility [5].

Delta Lake was preferred over plain Parquet or CSV because Parquet alone does not provide ACID transactions, schema enforcement or time-travel functionality. These three properties are essential for a pipeline designed to be re-runnable and auditable: schema enforcement acts as an automated quality gate between layers, ACID transactions prevent partially-updated states during interrupted writes, and time travel enables independent reproduction of any published result without explicit snapshotting.

The gold layer schema consists of: BoroughName (string), year (integer), total_crimes (integer), total_fly_tipping (integer), recycling_rate (double), crime_risk (string) and neglect_score (string).

1.6 Processing and Transformation

The data processing logic converts the raw CSV data to aggregated data for analysis. For the crime data, dates in 'yyyyMM' format were transformed to date data types, and zero values were filled in for missing crime counts. The data was reshaped from wide to long format using stack operation, and transposed month-year columns into rows. For the fly-tipping dataset, commas were removed from numeric strings and borough names standardised by removing 'COUNCIL' from names. Median was used to fill in missing values. For the recycling dataset, financial year columns were renamed and unpivoted to long format. The three data sets were grouped at the borough-year level and inner-joined.

An inner join was preferred to a left or outer join because the correlation and regression analysis requires complete observations across all three series simultaneously. Retaining borough-year rows where one series was absent would have propagated null values into the statistical calculations or required cross-source imputation, conflating measurement gaps with substantive zeros. The cost is a modest reduction in a panel coverage at the early temporal extremes; this trade-off is acceptable given that the study prioritizes robust cross-borough association over exhaustive longitudinal coverage.

1.7 Analytics Approach

Pearson correlation coefficients were used to determine the linear relationship between two variables. Multiple linear regression was used to determine the relationship between the dependent variable; total crimes and independent variables, number of fly-tipping incidents and recycling rates. The multiple linear regression equation can be written as: $\text{Crime} = \beta_0 + \beta_1(\text{Fly-tipping}) + \beta_2(\text{Recycling Rate}) + \varepsilon$. The R^2 statistic is a measure of how well the overall model explains variance in crime rates.

Pearson correlation was selected because the study aimed to measure linear relationships between continuous numerical variables. Preliminary visual inspection of the scatter plots in 3.3 confirmed an approximately linear association, supporting the parametric specification over rank-based alternatives such as Spearman correlation. The reported coefficients should therefore be read as estimates of linear co-variation rather than evidence of any causal mechanism.

Multiple linear regression was preferred to a univariate correlation framework because fly-tipping and recycling are conceptually distinct mechanisms whose independent contributions to crime should be

estimated jointly; univariate correlations leave open the possibility that one proxy merely confounds the other. More complex specifications such as fixed-effects panel models were considered but rejected because they would absorb the cross-borough variance the theory predicts, removing the signal the study is designed to detect.

Tableau was chosen as the visualization tool due to its ability to create interactive dashboards for exploring relationships. A range of visualizations were produced: scatter plot of crime versus fly-tipping, scatter plot of crime versus recycling rate, risk heat map of crime risk by neglect score, line chart of crime trends over time, bar chart of top boroughs by crime, and bar chart of top boroughs by fly-tipping.

Tableau is well-suited because the intended audience includes non-technical policy stakeholders who require an exploratory interface without coding expertise. Its interactive filtering, filter controls and built-in trend-line functionality support the hypothesis-testing and exploratory analysis central to the study's objectives. Gold-layer data was exported as CSV and loaded into Tableau, which provided sufficient performance for the dataset volume while avoiding the configuration overhead of a live Databricks SQL endpoint.

Part 2: Technical Build

2.1 Implementation Overview

The Microsoft Azure environment included an Azure Data Lake Storage Gen2 account, the main storage system, and an Azure Databricks workspace with a single-node cluster. Containers were structured to reflect the medallion architecture by creating bronze, silver and gold containers within a single storage container. The solution was structured in a single Databricks notebook with sequential execution cells following the linear pipeline model: configuration, data load, bronze layer persistence, silver layer transformations, gold layer aggregations and analytics. This facilitated independent execution of each stage, while enabling automated processing.

2.2 Data Engineering

The crime dataset needed the largest transformation as it was in wide format with separate columns for each month-year. An unpivot transformation using stack function in Spark transformed the dataset into long format, with one row per borough-month pair. This expanded the rows from 33 to about 5,000 rows allowing year-on-year comparisons. The fly-tipping data required string cleaning steps, to eliminate commas from strings and to remove 'COUNCIL' from borough names. The recycling dataset required column renaming and unpivoting of financial year columns.

Reshaping to long format was a structural prerequisite rather than a stylistic choice. The source crime dataset encodes each month-year as a separate column, which precludes both temporal aggregation and joining against the annually reported fly-tipping and recycling series. Long format collapses the temporal dimension into a single key field, enabling the downstream borough-year aggregation and inner join on a uniform composite key; without this transformation the pipeline cannot produce a single analytical table from the three sources.

The three datasets were aggregated to the borough-year level for joining. Crime was aggregated as annual count, fly-tipping was aggregated as annual count and recycling was aggregated as annual average. We took advantage of Delta Lake's schema enforcement in the silver layer to reject writes with unexpected schema, ensuring data quality for downstream analytics.

2.3 Automation and Pipeline

The pipeline was crafted to be idempotent whereby executing the same notebook twice results in the same output. All writes were performed with the "overwrite" mode, which overwrites existing data. Delta Lake's 'overwriteSchema' was used when schema changes were anticipated to avoid schema mismatch issues. The notebook used base_path variables to specify storage paths.

Idempotency was treated as a design requirement rather than a convenience property because reproducibility is a precondition of the analytical claim being made: any independent reviewer must be able to re-execute the pipeline and recover identical outputs. Overwrite mode eliminates the append-mode failure case in which a partial re-run silently doubles a subset of rows, which would corrupt the statistical results without triggering an error. The trade-off, loss of intermediate write history is mitigated by Delta Lake's time-travel capability and the immutability of the bronze layer.

The notebook was divided into sections: configuration cells storing storage account keys, bronze layer cells for reading CSVs and initial Delta writes, silver layer cells for data cleaning, gold layer cells for aggregations and joins, analysis cells for correlation and regression analysis with MLflow integration, and export cells for writing CSVs for Tableau. This allowed targeted re-execution of sections of the pipeline for development.

MLflow was used to log experiment parameters (feature set, dataset version), evaluation metrics (R^2 , correlation coefficients) and the trained regression model as an artefact, ensuring full reproducibility of the analytical results.

2.4 Data Quality Handling

Data were checked for missing values. The crime dataset had missing crime counts (replaced with zero, assuming unreported crime indicates no crime). The fly-tipping dataset had missing total_incidents in early years, which were imputed using medians for each borough to maintain statistics. The recycling dataset contained full data coverage.

Median rather than mean imputation was applied because the fly-tipping distribution is right-skewed by a small number of high-incident boroughs; the mean would have pulled imputed values toward these extremes, inflating apparent neglect in boroughs that genuinely lacked observations. Imputation was performed within borough rather than across the full panel to avoid suppressing the cross-borough variance the analysis is designed to detect.

Invalid entries were handled as follows. Numerical entries with commas needed replacement using regex to remove commas prior to typecasting, with regex pattern `'^[0-9,]+$'` to ensure numeric strings. Column names containing spaces, slashes and commas were renamed to Delta Lake-compatible format. Date parsing allowed for errors, filtering out invalid dates but retaining valid data.

Part 3: Analysis Report

3.1 Descriptive Analysis

Crime Trends Over Years

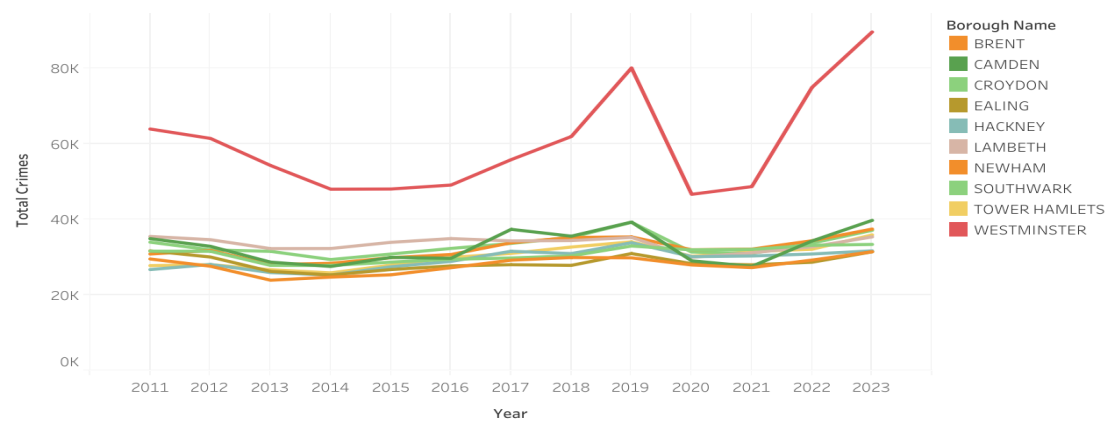


Figure 2: Annual Crime Trends by Borough (2011-2023)

Analysis of crime trends between 2011 and 2023 shows variation amongst London boroughs. Westminster consistently accounted for the largest number of crimes, recording an annual peak of approximately 89,000 offences in 2023. Boroughs like Lambeth, Southwark and Camden consistently reported high crime levels, with outlying boroughs at much lower levels. Boroughs with high crime levels displayed relatively consistent year-on-year trends.

Top 10 Boroughs by Crime

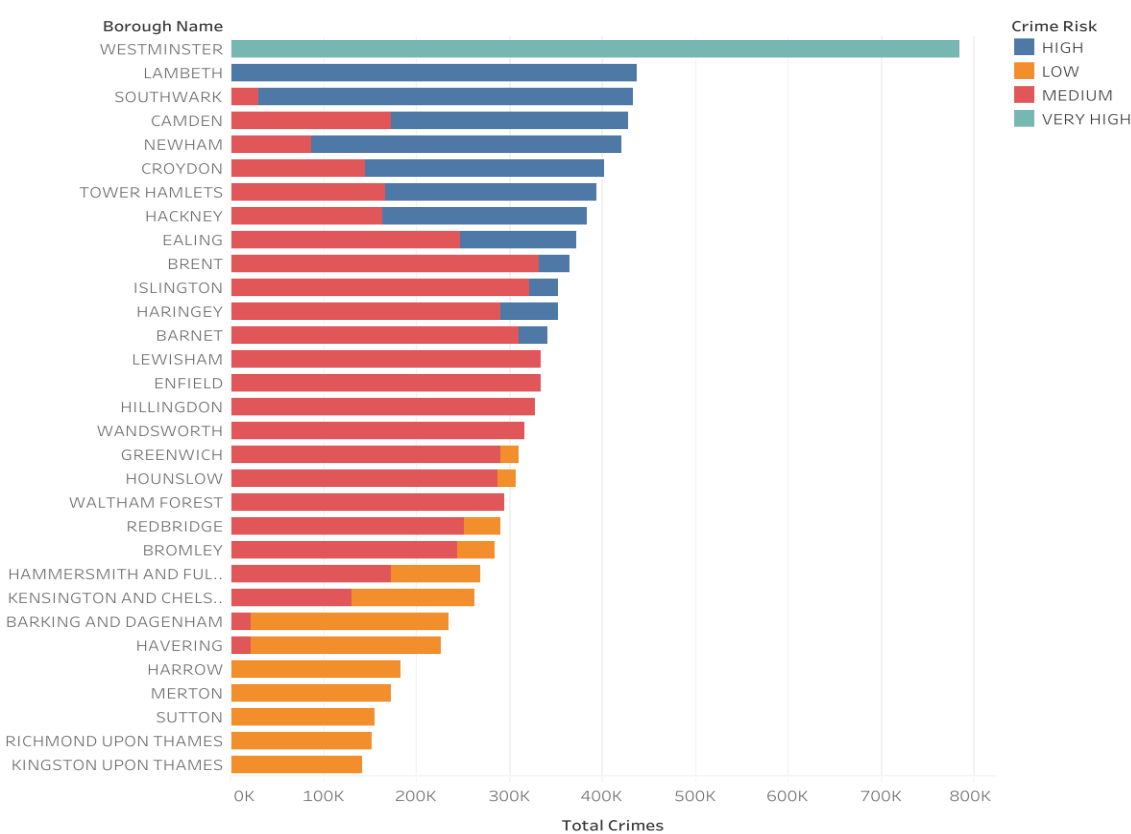


Figure 3: Cumulative Total Crimes by Borough with Crime Risk Classification

There is a high degree of geographical concentration. Westminster had the highest total crimes around 780,000 followed by Lambeth around 430,000 and Southwark around 420,000. Only Westminster was in very high-risk category and five in high or medium risk categories. The lowest scores were recorded by outer boroughs Sutton, Richmond and Kingston.

Top 10 Boroughs by Fly-tipping

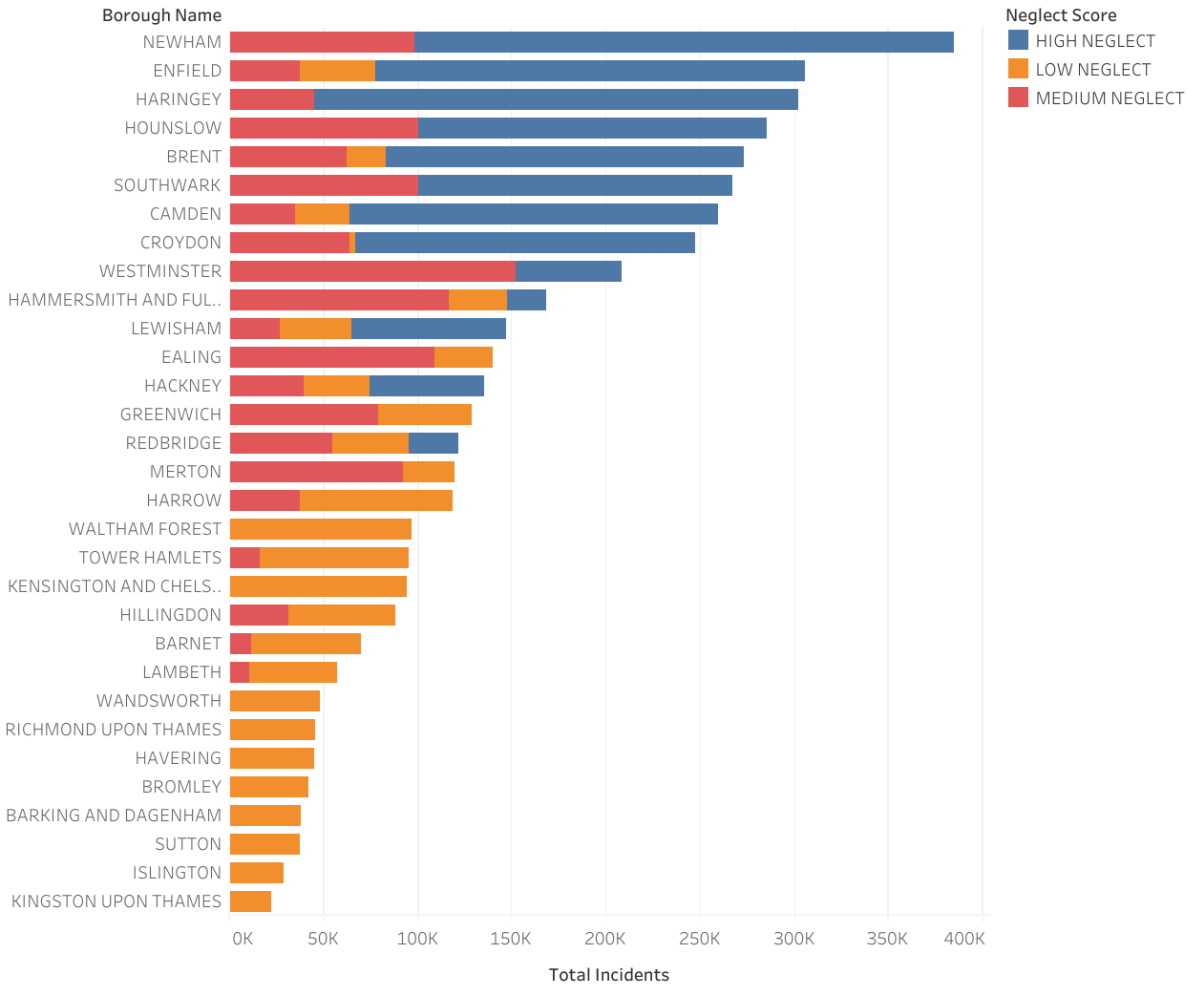


Figure 4: Cumulative Fly-Tipping Incidents by Borough with Neglect Score

The pattern of fly-tipping is different from crime. Newham and Enfield recorded the highest cumulative fly-tipping totals at approximately 390,000 and 270,000 incidents respectively over the study period. Rather than being in the centre of Westminster, fly-tipping was more likely to be in the east and north of London. This highlights that visible disorder might be affected by other factors than crime.

3.2 Statistical Analysis

Crime vs Fly-tipping

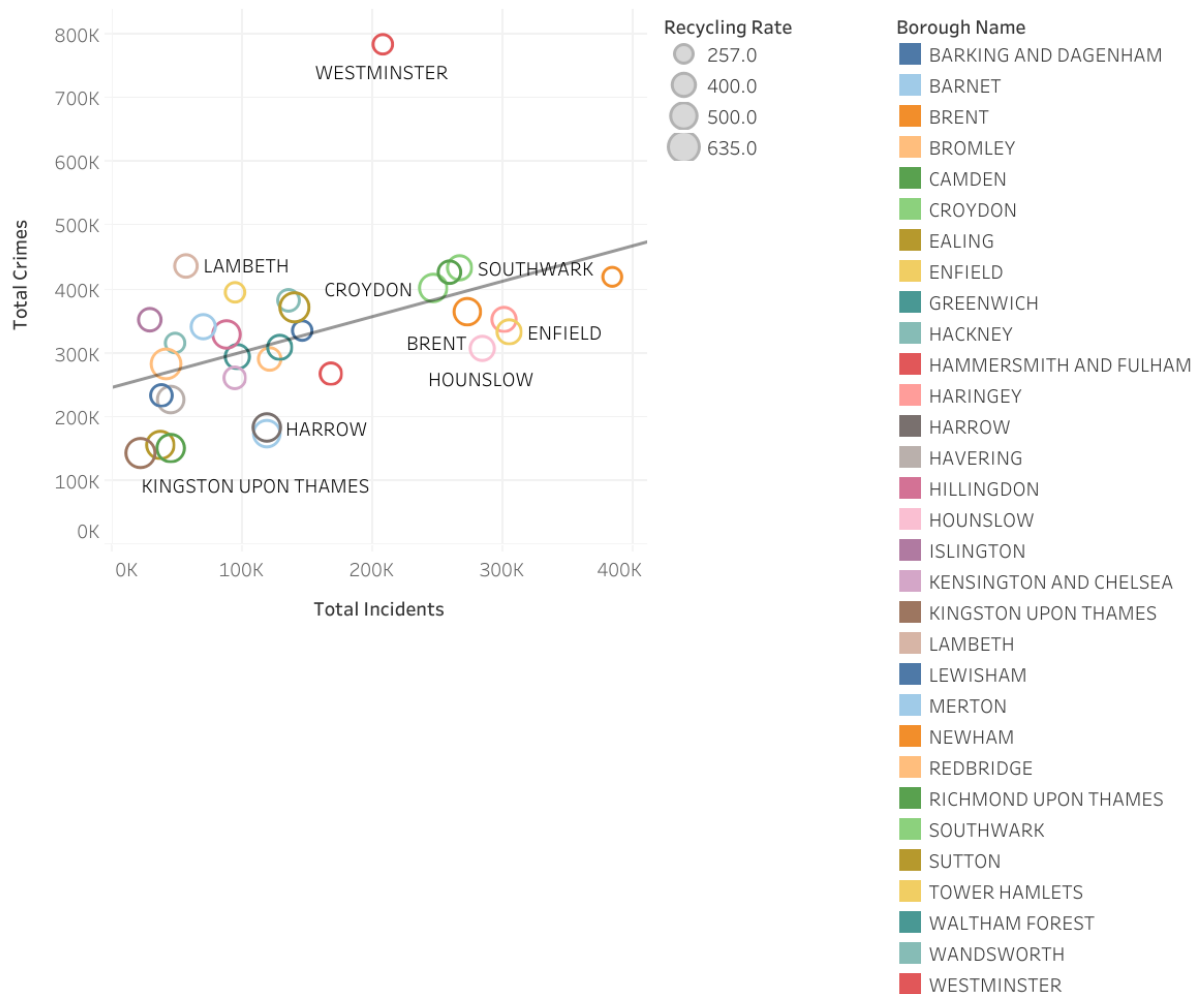


Figure 5: Scatter Plot of Total Crimes versus Fly-Tipping Incidents by Borough

A Pearson correlation test was conducted and a weak positive correlation of $r = 0.3185$ was found between crime and fly-tipping. Boroughs with greater rates of fly-tipping tend to have slightly greater rates of crime, offering some support to the visible neglect hypothesis. But the strength of the correlation indicates that fly-tipping only accounts for around 10% crime variation.

Both correlations were statistically significant ($p < 0.001$), confirming that neither association is attributable to chance at the 0.05 significance level.

Crime vs Recycling Rate

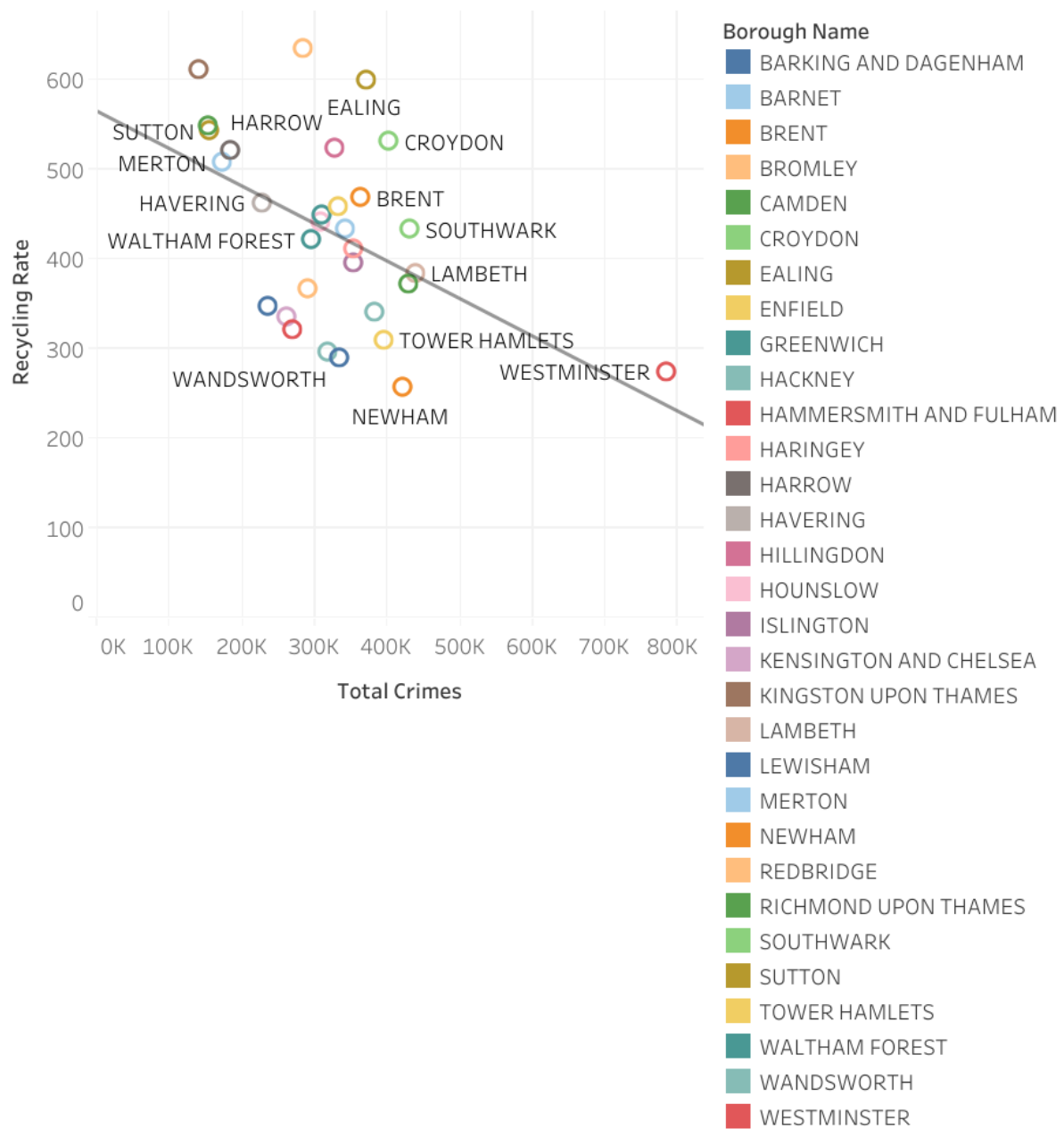


Figure 6: Scatter Plot of Total Crimes versus Recycling Rate by Borough

The correlation of crime and recycling rates was a moderate negative correlation of $r = -0.4302$. This is more significant than the fly-tipping correlation, implying that involvement in community engagement has a greater negative correlation with crime. Boroughs with higher recycling participation record lower aggregate crime totals.

This multiple linear regression (with both independent variables) produced an R^2 of 0.2517, suggesting 25% of the variation in crime can be predicted by fly-tipping and recycling. The regression equation is:

$$\text{Crime} = 0.23 \times \text{Fly_tipping} - 450.17 \times \text{Recycling_Rate} + 37,222.12$$

The negative sign on the recycling rate means for each 1% increase in recycling, crime decreases by 450 units. The low R^2 means considerable variance remains unexplained, and there are other factors besides environmental neglect that affect crime.

An R^2 of 0.252 should not be interpreted as evidence of a poor model in absolute terms; it must be evaluated against the complexity of the phenomenon being modelled. Crime at borough level is the joint outcome of socio-economic structure, policing intensity, demographic composition and built-environment design, among many other determinants. A two-predictor specification that accounts for approximately 25% of cross-borough variance is materially non-trivial and is broadly consistent with effect sizes reported in the meta-analytic Broken Windows literature [6].

The overall regression model was statistically significant ($F=67.29$, $p < 0.001$) with an adjusted R^2 of 0.248 based on 403 borough-year observations. The fly-tipping coefficient ($\beta=0.23$, $p < 0.001$, 95% CI [0.16, 0.31]) and recycling coefficient ($\beta= -450.17$, $p < 0.001$, 95% CI [-548.89, -351.44]) were both individually significant, confirming that each predictor contributes independently to the model.

3.3 Visual Insights

Crime Risk vs Neglect Score

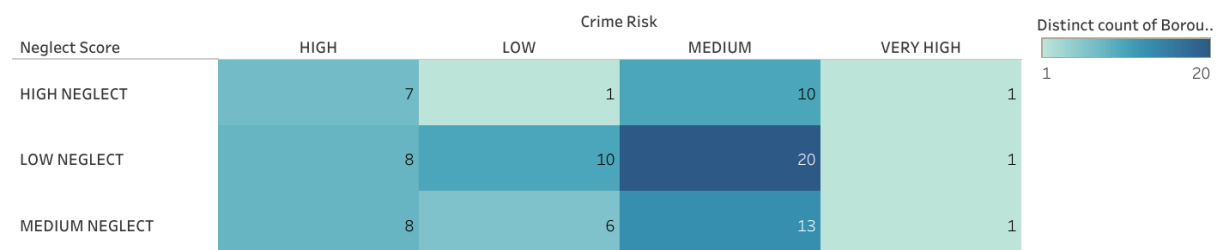


Figure 7: Heat Map of Crime Risk Category versus Neglect Score

The risk heat map shows that boroughs with high neglect are mostly found in medium and high crime risk categories, with 10 and 7 borough-year observations respectively. Boroughs with low neglect are most prevalent within the medium crime-risk category (20 observations), followed by the low crime-risk category (10 observations). Only 1 low-neglect borough-year observation appears in the very high crime category, suggesting that lower environmental neglect is generally associated with lower crime severity.

The line chart shows the high-crime boroughs did not converge with low-crime boroughs, and were consistent over time. The scatter plot of crime versus fly-tipping shows significant scatter around the regression line, suggesting many boroughs significantly differ from the predicted relationship. The scatter plot of crime versus recycling demonstrates a stronger negative association, with all boroughs recycling above 45% in the bottom half of the crime distribution.

3.4 Theory Evaluation

This study supports Broken Windows Theory in a complex way for modern London. The fly-tipping correlation provides weak to moderate support for the visible neglect hypothesis, in line with Wilson and Kelling's theory [1] that visible neglect will facilitate further crime. The recycling correlation offers solid evidence for the sense of community hypothesis, consistent with O'Brien et al's meta-analysis [6] which concluded there was greater evidence for community-based than disorder-crime hypotheses.

The R^2 value of 0.2517 suggests that 25% of the variation in crime is explained by environmental neglect and recycling. Classic Broken Windows Theory emphasizing visible signs of disorder may not explain London's crime. Other factors such as socio-economic background, police strategies, population density and the economic environment likely have a much greater influence on crime variance than environmental variables. As such, this study finds Broken Windows Theory is partially supported for London as directionality of effects is consistent with theory, but the strength of effects is weak.

The difference of strength between the two indicators provides an important theoretical elucidation. Active community participation, by recycling behavior, is more associated with crime than fly-tipping, passive environmental neglect. This indicates that positive community-level factors may be more important for reducing crime than the lack of negative factors. Community-based interventions that involve building community engagement and collective efficacy may be more effective than those targeted toward cleaning up visible disorder.

The discovery does have an impact on London policy makers. Initiatives that focus on community, such as neighborhood improvement schemes, encouraging recycling and environmental projects in local communities, can provide more crime reduction benefits than enforcement-based approaches that focus only on visible aspects of neglect. But the low R^2 values caution against just using one theory of crime prevention as a sole approach to crime prevention. A combination of economic inequality, policing effectiveness and social services along with environmental improvements would have the greatest impact in reducing crime over London boroughs.

3.5 Limitations

The results are subject to a number of limitations. Correlation is not causation, as this study cannot ascertain whether neglect leads to crime, or whether both neglect and crime are caused by another, unmeasured variable such as poverty. Borough-level aggregation obscures within-borough differences, with high- and low-crime areas within London boroughs. Few variables limit explanatory capacity, as many other variables are excluded such as unemployment, income inequality, and police numbers. Issues of validity undermine both independent variables, with fly-tipping data reflecting reporting behaviour as well as actual fly-tipping and recycling rates potentially reflecting the effectiveness of council policy rather than genuine community participation. Data was aggregated to the lowest common denominator of monthly, annual, and annual data, which may have obscured short-term dynamics.

3.6 Conclusion

This study has designed and implemented a cloud-based data lakehouse analytics pipeline to explore Broken Windows Theory in London. The testing of the theory supported it partially, with the weak positive correlation between fly-tipping and crime at $r = 0.3185$ and the moderate negative correlation between recycling and crime at $r = -0.4302$. But the multiple regression model only accounted for 25% of variance in crime with $R^2 = 0.2517$, suggesting environmental neglect and community participation are only weak predictors at the borough level.

The data pipeline worked well, ingesting three data sets with more than a million records, transforming them including unpivoting wide crime data and delivering data suitable for analysis. The medallion approach guaranteed quality bronze, silver and gold and Tableau reports allowed interactive visualizations of spatial-temporal variations in crime. The low R^2 value suggests policymakers should consider environmental interventions as part of a suite of crime prevention measures. Future research should include socio-economic variables and conduct finer spatial analyses to enhance model performance.

References

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AI Declaration

I used AI to assist with grammar improvement, wording refinement, troubleshooting small code issues and clarifying technical concepts during this assessment. The data pipeline, Databricks implementation, Tableau analysis, statistical interpretation and conclusions were completed and verified by me. AI was not used to generate citations or references, which were checked manually from original sources.